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JSAE Review 16 (1995) 145-150

Transferred load and its course in passenger car bodies

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Received 30 September 1994

Abstract

A method to express the concept of load transfer and the course of the transferred load in the structure is introduced. Transferred load in an actual passenger car body is investigated experimentally. The degree of joint stiffness between the members of the vehicle body can also be discussed based on this concept. Difference can be shown between the course from the front suspension and that from the rear suspension under the condition of the torsional loading.

1. Introduction

In designing the passenger car bodies, it is important to take the stiffness of body structures into consideration not only for the fundamental requirements of the structures but also for the requirement of the vehicle dynamics and stability. The stiffness of the body structures or the joint stiffness of the vehicle body members have been widely studied [1–3].

In this paper, the authors propose a new parameter E^* that expresses the degree of connection between a loading point and an arbitrary point in the structure. Parameter E^* is introduced based on the concept of relative stiffness [4].

Distribution of the parameter E^* in an actual vehicle body can be measured experimentally. From the measured distribution of E^*, the load transfer course in the body structure is obtained, and the internal continuity of stiffness can be examined.

Two load transfer courses in the body structure are measured under torsional loading. One is the course from the front suspension and the other is the course from the rear suspension.

2. Relative stiffness and parameter E^*

2.1. Introduction of parameter E^*

Figure 1(a) shows an elastic material which has three points A, B and C. A is the loading point, B the fixed end and C the arbitrary point in the elastic body. The elastic constant between the arbitrary two points may be called relative stiffness [4]. The load-displacement relation for an elastic body using relative stiffness can be expressed as follows:

$$\begin{bmatrix} \boldsymbol{p}_A \\ \boldsymbol{p}_B \\ \boldsymbol{p}_C \end{bmatrix} = \begin{bmatrix} \boldsymbol{K}_{AA} & \boldsymbol{K}_{AB} & \boldsymbol{K}_{AC} \\ \boldsymbol{K}_{BA} & \boldsymbol{K}_{BB} & \boldsymbol{K}_{BC} \\ \boldsymbol{K}_{CA} & \boldsymbol{K}_{CB} & \boldsymbol{K}_{CC} \end{bmatrix} \begin{bmatrix} \boldsymbol{d}_A \\ \boldsymbol{d}_B \\ \boldsymbol{d}_C \end{bmatrix}. \tag{1}$$

Each $\boldsymbol{K}$ with suffixes indicates the relative stiffness. $\boldsymbol{p}$ and $\boldsymbol{d}$ with suffixes show the force and displacement vectors respectively. The displacement $\boldsymbol{d}_B$ is vanished under the condition in Fig. 1(a):

$$\boldsymbol{d}_B = 0. \tag{2}$$

From Eqs. (1) and (2), the next relation

$$\boldsymbol{p}_A = \boldsymbol{K}_{AA}\boldsymbol{d}_A + \boldsymbol{K}_{AC}\boldsymbol{d}_C \tag{3}$$

can be obtained, and the condition of the invariance with a rigid translation may be described as follows:

$$\boldsymbol{K}_{AA} = -(\boldsymbol{K}_{AB} + \boldsymbol{K}_{AC}).$$

This equation shows the relation between three matrices which are concerned with the loading point A. Matrix $\boldsymbol{K}_{AA}$ may be used to express the degree of connection between A and C. Vector $\boldsymbol{p}_A$ has three components and $\boldsymbol{d}_A$ is identical, therefore $\boldsymbol{K}_{AA}$ has nine components.

On the other hand, the stored energy in the structure in Fig. 1(a) is given by the next relation:

$$E = \tfrac{1}{2}(\boldsymbol{p}_A)^{\mathrm{T}}\boldsymbol{d}_A \tag{4}$$

where, the superscript $(\ )^{\mathrm{T}}$ shows the transpose of a col-

SSDI 0389-4304(95)00010-0

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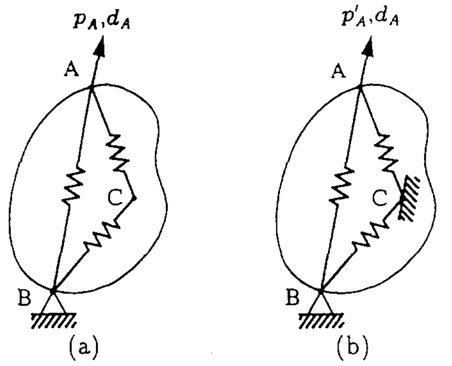

Fig. 1. Relative stiffness and E^*.

umn vector, and Eq. (4) is the scalar product of the vectors. The stored energy can be also expressed as follows with the use of Eq. (3):

$$E = \tfrac{1}{2}(\boldsymbol{K}_{AA}\boldsymbol{d}_A + \boldsymbol{K}_{AC}\boldsymbol{d}_C)^{\mathrm{T}}\boldsymbol{d}_A. \tag{5}$$

In the second loading case shown in Fig. 1(b), A is a loading point, B is a fixed point and the arbitrary point C are also fixed. The displacement at point A in the second loading case is given the same value as the displacement in the first loading case:

$$\boldsymbol{d}_{B'} = \boldsymbol{d}_{C'} = 0, \ \boldsymbol{d}_{A'} = \boldsymbol{d}_A. \tag{6}$$

In this paper, a letter with a prime ( )′ shows a value under the second condition. The external force [Eq. (3)] and the stored energy [Eq. (4)] may be rewritten as follows:

$$\boldsymbol{p}'_A = \boldsymbol{K}_{AA}\boldsymbol{d}_A, \tag{7}$$

$$E' = \tfrac{1}{2}(\boldsymbol{p}'_A)^{\mathrm{T}}\boldsymbol{d}_A = \tfrac{1}{2}(\boldsymbol{K}_{AA}\boldsymbol{d}_A)^{\mathrm{T}}\boldsymbol{d}_A. \tag{8}$$

It should be noted that $\boldsymbol{p}'_A \neq \boldsymbol{p}_A$. Equation (8) shows E' and $\boldsymbol{K}_{AA}$ are related directly, so that E' may also be used for the index to express the degree of connection between points A and C. In contrast with $\boldsymbol{K}_{AA}$, parameter E' is a scalar valued function which is desirable for simplicity. We rewrite Eq. (8) in the following nondimensional form:

$$E^* \equiv \frac{E'}{E} = \frac{(\boldsymbol{p}'_A)^{\mathrm{T}}\boldsymbol{d}_A}{(\boldsymbol{p}_A)^{\mathrm{T}}\boldsymbol{d}_A} = \frac{(\boldsymbol{K}_{AA}\boldsymbol{d}_A)^{\mathrm{T}}\boldsymbol{d}_A}{(\boldsymbol{K}_{AA}\boldsymbol{d}_A + \boldsymbol{K}_{AC}\boldsymbol{d}_C)^{\mathrm{T}}\boldsymbol{d}_A}. \tag{9}$$

Parameter E^* is the ratio of the elastic energy between the two cases of fixation for point C. The value of the parameter E^* can be defined at an arbitrary point C, and the distribution of E^* can be obtained for the whole structure.

Even though a part of the structure is sufficiently stiff, it is not significant for the total stiffness when the degree of connection with the loading point is insufficient. A stress value is certainly an effective index for failure, but is not an index for stiffness. It should be noted that the stress concentration has a large effect for failure, but has a little effect regarding total stiffness. On the contrary, E^* may be an effective index for the total stiffness of a structure.

2.2. Load transfer course

Through the above discussions it has become clear that stress distribution can never indicate the state of a load transfer and that the applied load may be transferred according to the values of E^*.

It may be reasonable to define the load transfer course as the line which connects the highest point of E^* successively. The load transfer course can be obtained as the curve which corresponds to the ridge line of the contour lines of E^*.

2.3. Internal continuity of stiffness along the course of load transfer

The concept of continuity can be shown with the examples shown in Fig. 2(a). Model 1 in Fig. 2(a) shows a cantilever with a concentrated force $\boldsymbol{p}$, and the contour lines of E^* are also displayed in the figure. Values of E^* are calculated by the finite element method and the contour lines are obtained mathematically with the spline function. The ridge line of the contour lines with coordinate s is the load transfer course defined above. Figure 2(b) shows the distribution of E^* along the ridge line. Abscissa s in Fig. 2(b) is non-dimensionalized by the total

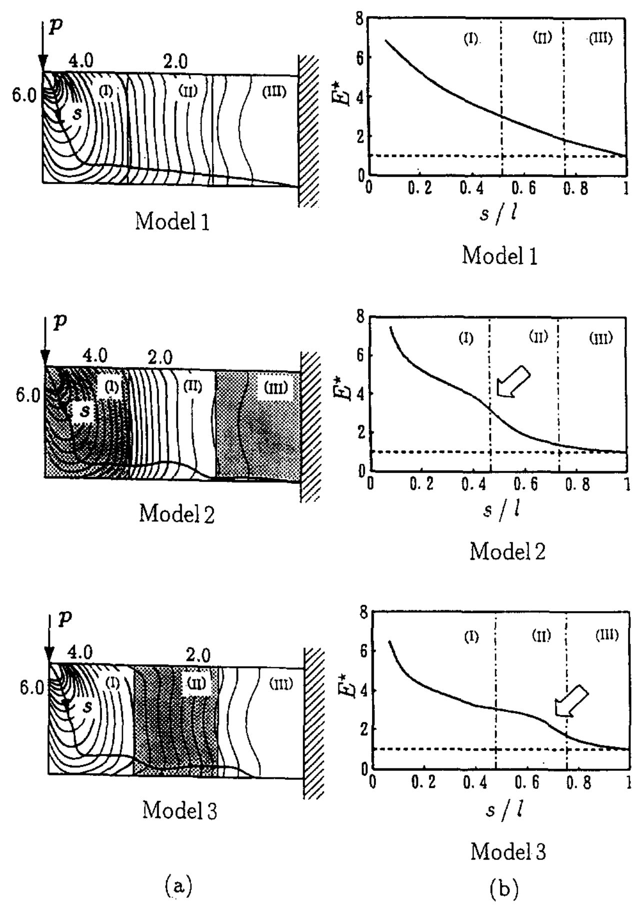

Fig. 2. Load transfer course and continuity of stiffness.

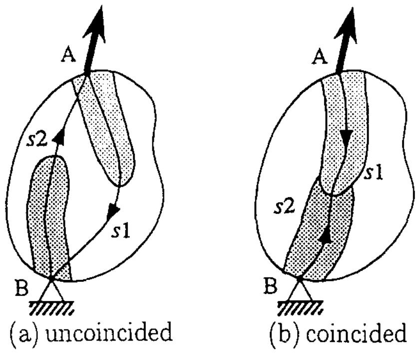

Fig. 3. Coincidence of load transfer courses ($s1$: from A to B, $s2$: from B to A).

length of the ridge line l. Model 1 is a cantilever with a uniform thickness, while in the regions (I) and (III) of Model 2 or region (II) of Model 3, that is in the shaded parts, thicknesses are twice as thick as Model 1.

Figure 2(b) shows that E^* of Model 1 decreases smoothly along coordinate s. Whereas, in Model 2 and 3, E^* decreases suddenly at the marked points with the arrows, and it means that the degree of connection with the loading point decreases at the point suddenly. These curves show that the Model 1 is more desirable than Model 2 or 3 from the standpoint of internal continuity of stiffness. It can be concluded that E^* may be an effective index to discuss the continuity of stiffness in structures.

In this paper, continuity of stiffness does not mean the mathematical continuity, but it means the successive conjunction of stiff members in a structure.

2.4. *Coincidence of load transfer courses*

The degree of connection E^* at an arbitrary point can be defined by two cases. One is the degree of connection with a loading point, and the other is that with a fixed end. Corresponding to these two E^*s, two courses of load transfer can be obtained. Relation between these two courses should be also examined. In Figs. 3(a) and 3(b), the course from the loading point is shown as curve $s1$, and the other as curve $s2$.

If these two courses do not coincide as in Fig. 3(a), the state of load transfer is not effective and the two courses should be connected by other stiff members. It is desirable that these two courses coincide as shown in Fig. 3(b).

3. Measurement of E^* in an actual vehicle body

In order to obtain the load transfer courses and to discuss the internal continuity of stiffness, values of the parameter E^* are measured experimentally in an actual vehicle body. The vehicle body whose roof panel has especially low stiffness is investigated under the torsional loading. Values of E^* are measured in two loading cases as shown in Fig. 4 and 5. Reaction forces at the supported ends are shown in Figs. 4(b) and 5(b) schematically. Considering the free body diagram, we can recognize that both loading cases are identical and the pure torsion is applied to the body in both cases. An arbitrary point in the body is fixed and the value of applied force is measured to obtain the value of E^*. Loading is vibrational to avoid the effect of friction. The degree of connection with the front end is measured by the loading condition in Fig. 4, and the degree of connection with the rear end is measured by the condition in Fig. 5. Detailed method to obtain E^* is as follows.

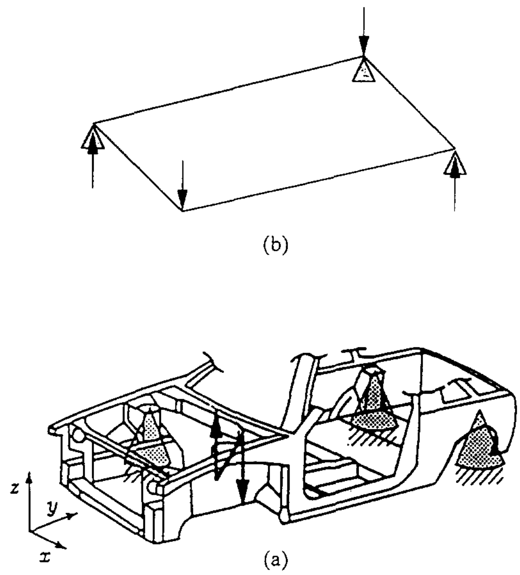

Fig. 4. Torsional loading condition 1.

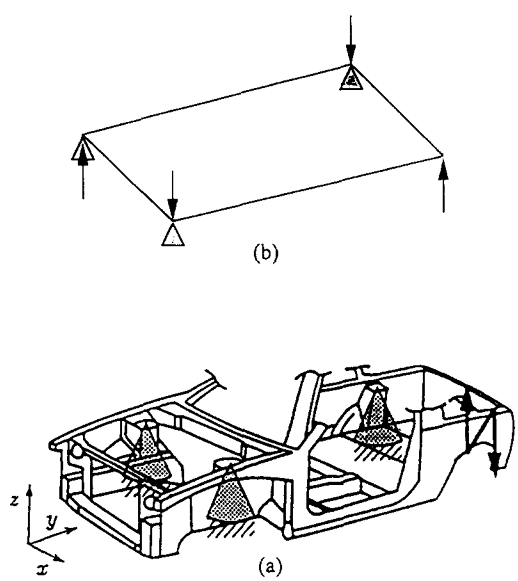

Fig. 5. Torsional loading condition 2.

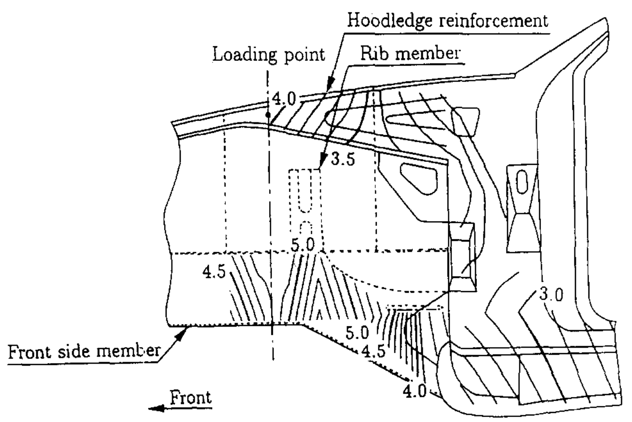

Fig. 6. Distribution of E^* at front end structure (torsional loading condition 1).

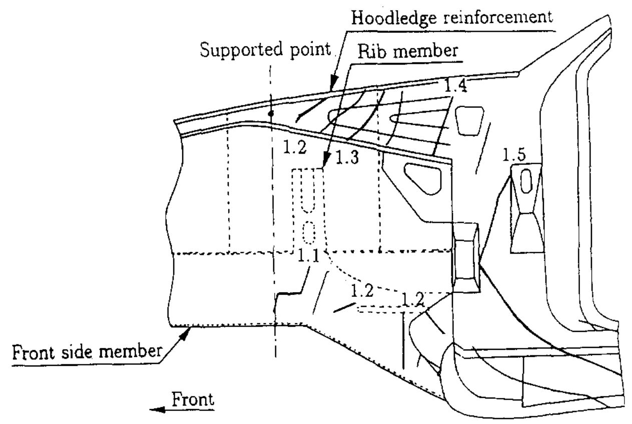

Fig. 7. Distribution of E^* at front end structure (torsional loading condition 2).

Equation (4) can be simplified by the condition in which the x, y, z components of the displacement $\boldsymbol{d}_A$ at the loading point are $(0, 0, d_{zA})$.

$$E = \tfrac{1}{2} p_{zA} d_{zA} \tag{10}$$

where p_{zA} and d_{zA} show z components of $\boldsymbol{p}_A$ and $\boldsymbol{d}_A$. Similarly, Eqs. (8) and (9) are also simplified:

$$E' = \tfrac{1}{2} p'_{zA} d_{zA}, \tag{11}$$

$$E^* \equiv \frac{E'}{E} = \frac{p'_{zA}}{p_{zA}}. \tag{12}$$

To obtain E^* in Eq. (12), we have to measure external force p'_{zA} when the measure point is fixed, and the force p_{zA} when the measure point is free.

4. Results of experiment

4.1. Distribution of E^* in the vehicle body

4.1.1. Distribution of E^* at front end structure

Figure 6 shows the distribution of E^* at the front end structure in the loading condition in Fig. 4. Values of E^* in Fig. 6 show the degree of connection between each point and the front suspension. It can be shown that the values of E^* at the front side member are higher than E^* at the hoodledge reinforcement. The maximum value of E^* can be seen at the center of the front side member which is just under the rib member. Therefore, it can be concluded that the external load is transferred mainly to the front side member through the rib member.

Figure 7 shows the distribution of E^* at the front end structure in the loading condition in Fig. 5. Values of E^*

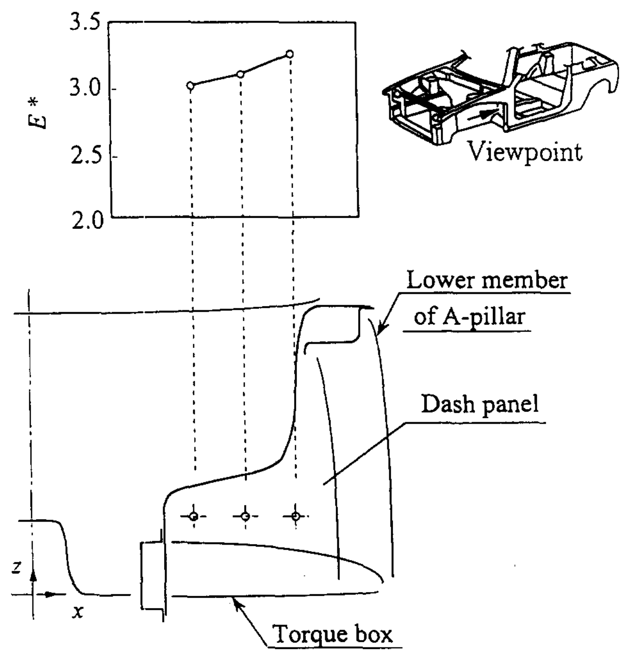

Fig. 8. Distribution of E^* at dash panel (torsional loading condition 1).

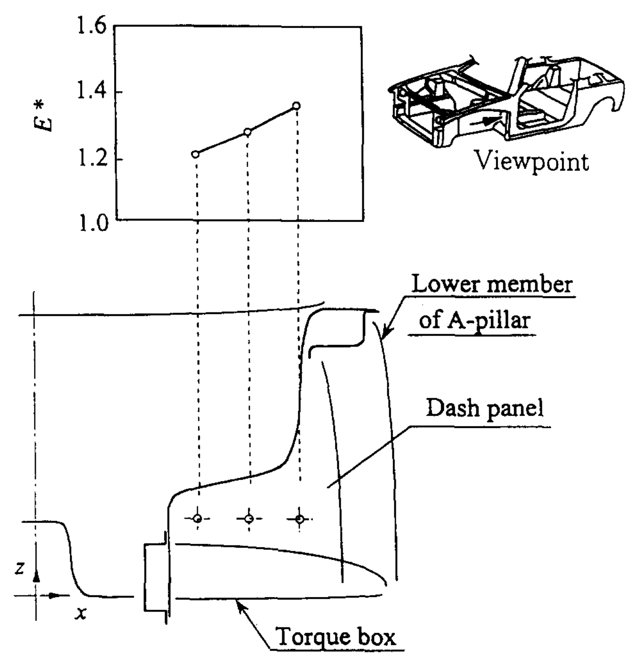

Fig. 9. Distribution of E^* at dash panel (torsional loading condition 2).

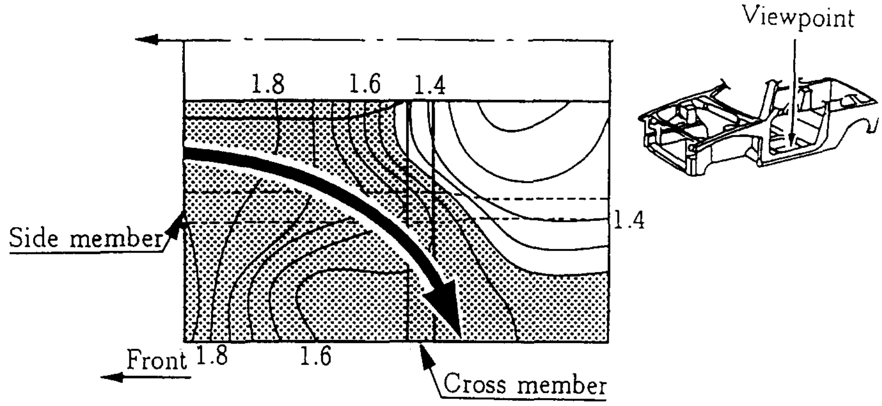

Fig. 10. Distribution of E^* at floor panel (torsional loading condition 1).

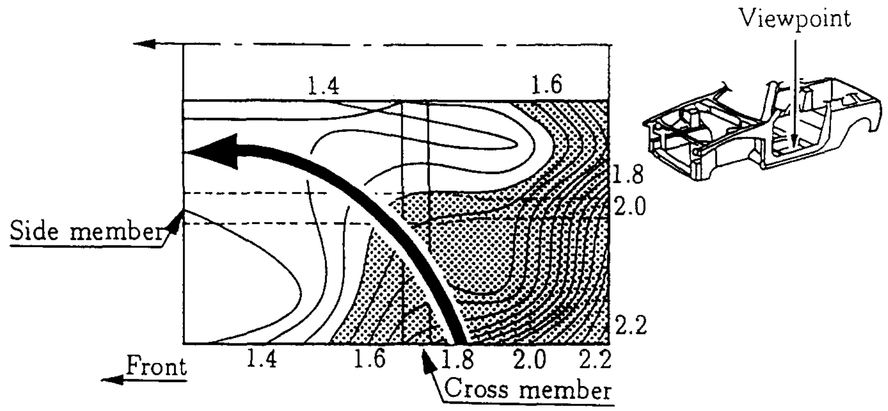

Fig. 11. Distribution of E^* at floor panel (torsional loading condition 2).

in Fig. 7 show the degree of connection between each point and the rear suspension. It can be shown that the values of E^* at the hoodledge reinforcement are higher than E^* at the front side member, which means that the hoodledge reinforcement and the rear suspension are connected strongly to each other. Therefore, it can be said that the external force at the rear suspension is transferred mainly to the hoodledge reinforcement through the lower member of the A-pillar.

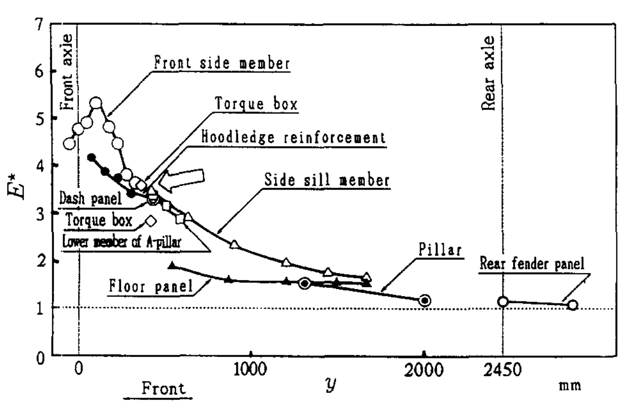

Fig. 12. Distribution curve of E^* for each member (torsional loading condition 1).

Comparing Figs. 6 and 7, it can be concluded that the course of the load transfer from the front suspension and the course from the rear suspension do not coincide as shown in Fig. 3(a). It is desirable that these two courses coincide as shown in Fig. 3(b).

4.1.2. Distribution of E^* at the dash board panel

Figure 8 shows the distribution of E^* at the dash board panel, which is in the case of loading at the front suspension. The value of E^* is higher at the outside of the body. It can be shown that the external force from the front suspension is transferred to the dash board panel through the lower member of the A-pillar.

Figure 9 shows the distribution of E^* at the dash board panel, which is measured in the loading at the rear suspen-

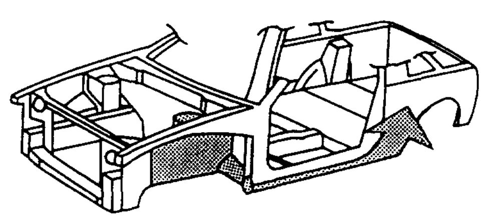

Fig. 13. Load transfer course (torsional loading condition 1).

sion. The value of E^* is also higher at the outside of the body. It shows that the external force from the rear suspension is also transferred to the dash board panel through the lower member of the A-pillar.

4.1.3. Distribution of E^* at floor panel

Figure 10 shows the distribution of E^* at the floor panel, which is in the case of loading at the front suspension. The load is transferred from the frontal part of the floor panel to the side sill member through the cross member as shown by the bold arrow in Fig. 10. Rear part of the side member and the tunneled part of the floor panel has a slight role regarding the load transfer.

Figure 11 shows the distribution of E^* at the floor panel, which is in the case of rear loading. The load is transferred from the side sill member to the frontal part of the floor panel as shown by the bold arrow in Fig. 11.

Comparing Figs. 10 and 11, it can be concluded that the course of the load transfer from the front suspension and the course from the rear suspension coincide in the floor panel as shown in Fig. 3(b).

4.2. Load transfer course in the whole body

Figure 12 shows the distribution curves of E^* for each member along the y coordinate throughout the whole body with the front loading. The external load in the front suspension is transferred mainly to the front side member. Subsequently, the load is transferred to the side sill member through the torque box. At the point marked with an bold arrow in Fig. 12, E^* decreases suddenly. It means that the degree of connection decreases at the point with the arrow suddenly. The course of the load transfer from the front suspension may be expressed as in Fig. 13.

Figure 14 shows the distribution of E^* along the y coordinate throughout the whole body under the rear loading case. The external load in the rear suspension is transferred to the rear fender panel. Subsequently, the load is transferred to the side sill member, the lower member of

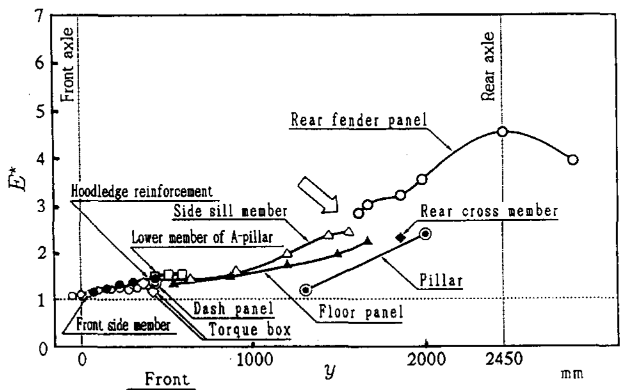

Fig. 14. Distribution curve of E^* for each member (torsional loading condition 2).

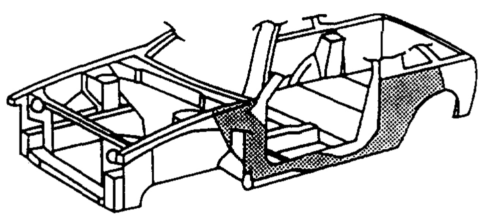

Fig. 15. Load transfer course (torsional loading condition 2).

the A-pillar and the hoodledge reinforcement. At the points marked with an bold arrow in Fig. 14, the degree of connection decreases suddenly. The course of the load transfer from the rear suspension may be expressed as in Fig. 15. Though the courses shown in Figs. 13 and 15 are slightly different, the coincidence of the courses in the whole body is almost satisfactory.

5. Conclusions

A method to express the concept of load transfer in the structure of a vehicle body is introduced. The degree of connection between the members of the vehicle body can also be expressed by the parameter E^*. The results of the experiment for an actual passenger car body are as follows:

(1) Improvement of the joint stiffness between

(1) the torque box and the side sill member,

(2) the side sill member and the rear fender panel

may be effective for the torsional stiffness of the body structure.

(2) A lack of coincidence can be seen between the course from the front suspension and the course from the rear suspension under the condition of the torsional loading.

Acknowledgments

The authors would like to express their gratitude to Mr. Kenichi Fukada and Dr. Kazuyuki Shizawa for their helpful discussions.

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